

Lat: **7.509N** Lon: **99.617E** Elev: **67** StdP: **14.66** Time Zone: **7.00 (E07)** Period: **94-19** WBAN: **99999**

TRANG, Thailand

WMO: **485670**

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	71.3	72.6	65.5	94.3	85.9	66.8	98.7	84.4	17.8	85.7	16.2	86.0	0.3	30	0.312

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	17.3	96.0	78.3	94.7	78.2	93.3	78.4	81.8	89.6	81.1	88.9	80.8	88.5	7.0	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.9	155.3	85.2	79.2	151.7	84.8	78.8	150.0	84.6	45.6	90.1	44.9	89.4	44.4	88.8	86.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
							Min	Max	Min	Max	Min	Max	Min	Max	
1%	2.5%	5%	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
16.2	14.1	12.2	DB	68.6	97.6	2.2	1.8	67.1	98.9	65.8	100.0	64.6	101.0	63.0	102.3
			WB	67.0	83.8	2.0	1.1	65.5	84.5	64.3	85.2	63.2	85.8	61.7	86.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	82.6	81.3	82.6	84.0	84.8	84.0	83.1	82.5	82.4	82.2	81.8	81.3	80.8
	DBStd	2.30	1.88	1.75	1.81	2.04	2.07	1.98	1.92	1.99	1.96	1.85	2.23	2.08
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11884	970	912	1053	1045	1055	992	1006	1006	965	987	939	954
	CDD65	6409	505	492	588	595	590	542	541	541	515	522	489	489
	CDH74	64315	4954	5325	6635	6578	6343	5635	5426	5362	4821	4554	4319	4363
	CDH80	25236	1845	2344	3187	3068	2630	2195	2057	1991	1698	1480	1362	1379
(14) Wind	WSAvg	3.4	5.2	5.5	4.0	2.6	2.4	2.8	2.8	3.0	2.7	2.4	3.1	4.6
(15) Precipitation	PrecAvg	89.5	2.5	1.3	1.9	4.1	10.4	9.4	9.9	10.4	14.1	11.2	9.5	4.9
	PrecMax	115.1	18.3	3.7	10.3	10.5	20.6	21.9	20.7	19.8	20.7	18.3	16.9	11.3
	PrecMin	69.9	0.0	0.0	0.0	0.1	2.2	3.3	3.6	5.0	8.0	5.1	3.0	0.4
	PrecStd	13.1	4.8	1.3	2.4	2.5	4.3	4.4	4.1	3.9	3.9	3.9	3.5	3.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.6	95.2	97.4	98.8	96.9	93.6	93.1	93.0	92.9	92.8	92.2	91.6
		MCWB	77.3	77.0	77.1	78.6	80.2	79.6	79.4	79.1	79.3	79.0	78.8	77.7
	2%	DB	90.9	93.4	95.4	96.5	94.3	92.5	91.6	91.6	91.4	91.0	90.7	90.0
		MCWB	76.7	76.4	76.9	78.9	80.2	79.6	79.1	78.9	78.9	78.9	78.6	77.3
	5%	DB	89.6	91.9	94.3	94.7	92.5	91.3	90.5	90.5	89.9	89.4	89.3	88.9
		MCWB	76.1	76.1	76.9	79.1	80.2	79.5	79.0	78.9	78.8	78.8	78.4	77.1
	10%	DB	88.2	90.7	92.8	92.9	90.8	89.7	89.2	89.1	88.2	87.6	87.5	87.2
		MCWB	75.9	75.9	77.0	79.2	80.0	79.5	79.1	78.9	79.0	78.8	78.3	76.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.4	79.8	80.8	82.3	82.6	82.2	81.6	81.3	81.4	81.0	80.8	80.0
		MCDB	87.7	89.9	90.6	92.0	90.7	89.1	88.5	88.4	88.5	87.9	87.7	87.0
	2%	WB	78.4	78.6	79.9	81.3	81.7	81.4	80.8	80.6	80.4	80.1	79.9	79.0
		MCDB	87.1	88.9	89.6	90.8	89.7	88.5	87.9	87.6	87.4	87.0	86.9	86.3
	5%	WB	77.6	77.8	79.1	80.6	81.0	80.8	80.2	80.0	79.8	79.6	79.2	78.3
		MCDB	85.9	87.8	88.9	90.0	89.0	88.0	87.1	86.8	86.6	86.4	86.1	85.2
	10%	WB	77.0	77.1	78.4	79.9	80.5	80.1	79.7	79.4	79.2	79.0	78.7	77.6
		MCDB	85.0	86.8	87.8	88.8	88.3	87.0	86.4	86.1	85.8	85.6	85.4	84.0
(35) Mean Daily Temperature Range	5% DB	MDBR	14.6	17.3	18.4	17.3	14.5	13.6	13.5	13.2	13.0	12.8	12.0	12.3
		MCDBR	16.6	19.4	20.8	19.7	16.9	15.8	15.5	15.4	15.3	15.2	15.2	15.6
		MCWBR	5.1	5.6	5.5	5.3	5.3	5.4	5.3	5.0	5.1	5.0	5.0	5.2
	5% WB	MCDBR	14.6	17.6	18.3	17.4	14.9	14.2	14.0	13.8	13.8	13.7	13.4	12.9
		MCWBR	5.1	5.7	5.8	5.7	5.5	5.6	5.5	5.3	5.3	5.2	5.1	5.0
(40) Clear-Sky Solar Irradiance	taub		0.469	0.478	0.487	0.473	0.452	0.457	0.460	0.455	0.445	0.447	0.457	0.465
	taud		2.258	2.209	2.219	2.263	2.353	2.368	2.345	2.345	2.370	2.364	2.326	2.287
	Ebn at Noon		269	270	268	267	267	263	263	269	275	275	270	267
	Edh at Noon		44	47	47	45	40	39	40	41	40	40	41	42
(44) All-Sky Solar Radiation	RadAvg		1572	1867	1881	1818	1612	1548	1528	1568	1516	1442	1327	1317
(45)	RadStd		128	116	177	102	92	92	76	59	51	105	96	135

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	+0.99	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (20 neighbors)	+0.34	+0.45	N/A	N/A	+0.34	+0.38	N/A	N/A	+133	+135